

Catherine Varas

📍 Sydney, Australia ✉ catherine.varas.padilla@gmail.com

🌐 [linkedin.com/in/catherine-varas](https://www.linkedin.com/in/catherine-varas)

🔄 [catherinevaraspadilla-commits](https://github.com/catherinevaraspadilla-commits)

🌐 catherine-varas.cofoundy.dev

Experience

One51 — Data & Analytics Consultancy

Data Engineering Intern

Sydney, Australia

Jul 2025 – Present

- Developed Python scripts and interactive Streamlit application for QlikView-to-Power BI migration metadata analysis, enabling data-driven platform transition decisions
- Implemented RAG framework in Azure AI Foundry for automated T-SQL translation, reducing manual SQL conversion effort by 80%+
- Built metadata visualization dashboards serving cross-functional stakeholders across 3+ concurrent migration projects

BBVA Perú — One of Latin America's Largest Banks

Associate Data Engineer

Lima, Peru

Jun 2024 – Jan 2025

- Led migration of millions of contractual records from IBM Mainframe to cloud infrastructure, ensuring zero data loss across 5+ banking product lines
- Engineered Apache Spark transformation pipelines with Control-M orchestration, processing large-scale batch workloads for regulatory compliance
- Designed and implemented data quality frameworks validating banking records against compliance standards, reducing data discrepancies by 40%+
- Developed Java backend routines for Mainframe/Oracle coexistence model, bridging legacy and modern systems for 200+ internal users
- Managed centralized contracts module across multiple squads, coordinating database schema and backend service delivery

BBVA Perú

Junior Data Engineer

Lima, Peru

Oct 2022 – Jun 2024

- Built Spark data pipelines for banking product data migration to cloud, processing and transforming records into optimized Parquet format
- Deployed and maintained 70+ components including Java backend services, Spark batch jobs, and database scripts across production environments
- Created reusable Spark modules for data cleaning, validation, and loading, reducing pipeline development time for new data sources by 50%+
- Automated parameterized data quality checks via Spark, eliminating manual validation across multiple product datasets
- Collaborated with international engineering teams across Colombia, Mexico, and Spain on shared banking data platform initiatives

Kyndryl

Cloud Computing Intern

Lima, Peru

Feb 2022 – Oct 2022

- Conducted disaster recovery architectural analysis for enterprise clients, evaluating failover strategies across Azure cloud infrastructure
- Contributed to Data Engineering community, sharing best practices and documentation for cloud migration patterns

Vida Software

Business Intelligence Analyst

Lima, Peru

Jul 2021 – Feb 2022

- Processed Big Data workloads and built retail performance reporting pipelines, delivering insights for 10+ retail client accounts
- Developed Hasura GraphQL-to-SQL Server integration pipelines, automating data flow between frontend and backend systems
- Designed Power BI dashboards for retail clients, enabling real-time KPI monitoring and data-driven decision making

- Built Python automation scripts reducing manual data processing tasks by 60%+

Projects & Achievements

T-SQL to Snowflake RAG Translator <i>Lead Developer</i>	Python, RAG, LLM, Snowflake 2026
• Architected AI-powered engine transforming legacy T-SQL into modern Snowflake SQL using RAG pipelines, LLM reasoning, and deterministic validators	
YOLO + SAM2 Lab Tracking <i>Developer</i>	Python, YOLOv8, SAM2, Computer Vision 2026
• Built automatic multi-rat tracking system combining YOLOv8 real-time object detection with SAM2 high-accuracy segmentation for laboratory research	
Traffic Big Data Analysis — Sydney <i>Developer</i>	Spark, HDFS, Docker 2025
• Analyzed Sydney traffic data using Apache Spark and HDFS in containerized environments for large-scale distributed processing	
Harvest-Jira Automation Engine <i>Developer</i>	Python, REST API 2025
• Engineered API synchronization tool between Harvest time-tracking and Jira, automating project management workflows across teams	

Education

University of Queensland <i>Master in Data Science</i>	Brisbane, Australia Feb 2025 – Jun 2026
• Coursework: Cloud Computing (Kubernetes, Docker), Machine Learning, Artificial Intelligence, Data Analytics	
Peruvian University of Applied Sciences (UPC) <i>Bachelor of Industrial Engineering</i>	Lima, Peru 2019 – 2023
• Focus: Project Management, Process Optimization, Systems Design	
Institute Cibertec <i>Associate's in Computer Science</i>	Lima, Peru 2019 – 2022
• Coursework: SQL, Java, Python, C#, Oracle, Web Development	

Skills & Additional

Data Engineering:	Java, Python, Apache Spark, SQL, ETL Pipelines, Databricks, Control-M
Cloud & DevOps:	Azure (Data Factory, Fabric, AI Foundry), AWS, Docker, Kubernetes, CI/CD
AI & ML:	RAG, LLMs, Spark MLlib, scikit-learn, MLOps, Prompt Engineering, Generative AI
Visualization:	Power BI, Streamlit, Tableau, Snowflake
Databases:	PostgreSQL, Oracle, MySQL, IBM Mainframe, SQL Server
Certifications:	Microsoft Fabric Data Engineer (DP-700), Fabric Analytics Engineer (DP-600), Power BI Data Analyst (PL-300), AWS Cloud Practitioner, Azure Fundamentals (AZ-900), Generative AI Fundamentals (Databricks)
Languages:	English (Professional), Spanish (Native)